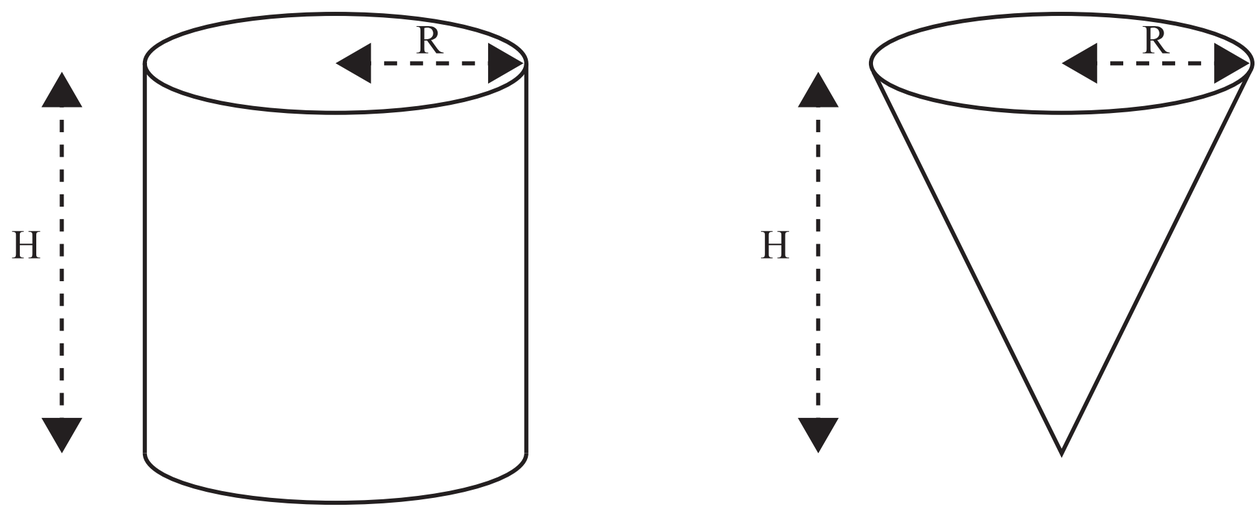
(e) The energy required to pump water out of a tank with a circular cross-section and height H is given by:

$$E = \int_0^{\mathrm{H}} \mathrm{k}(\mathrm{H} - h)A(h)\mathrm{d}h$$

where k is a constant
h is the height of the water in the tank at any instant
r is the radius of the water surface at this instant
$A(h)$ is the area of the surface of water at this instant.

A cylindrical tank and a conical tank are both full of water.

Both have height H, and the radius at the top of both tanks is R.



Show that the energy required to empty the conical tank is one sixth the energy required to empty the cylindrical tank.

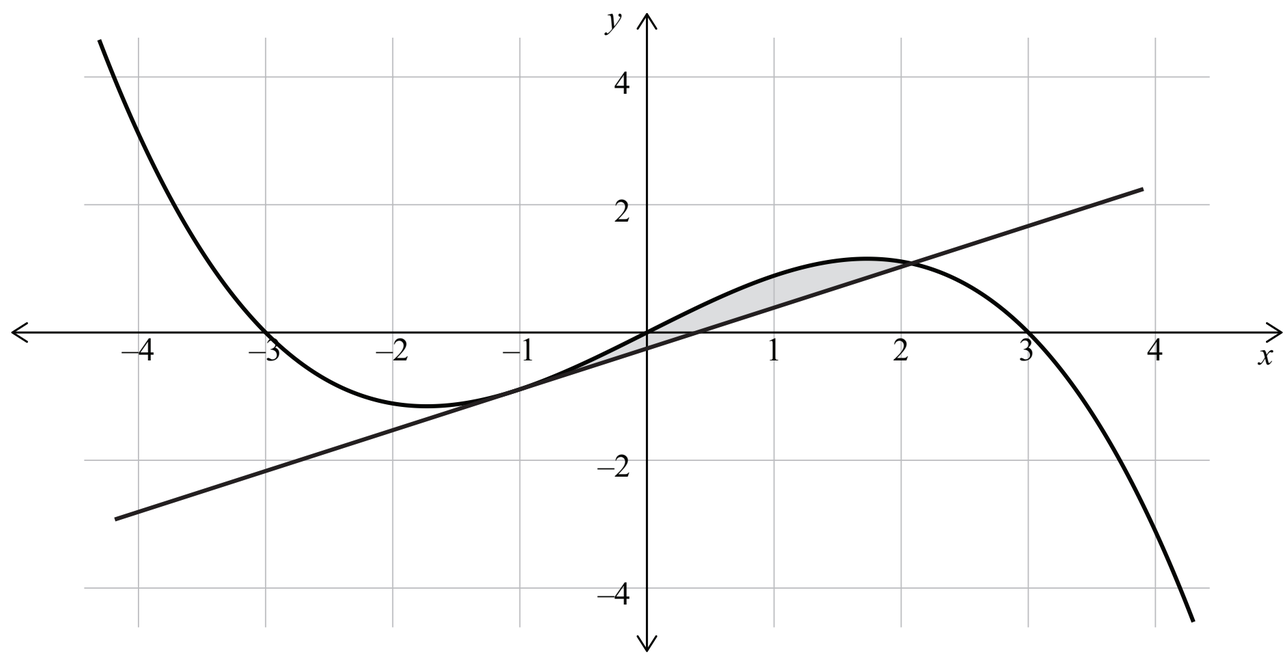
Give the results of any integration needed to solve this problem.

(e) A curve $y = f(x)$, which passes through the origin, is shown on the graph below.

Its gradient at any point is given by the equation $f'(x) = 1 - \frac{1}{3}x^2$.

The line on the graph is the tangent to the curve at $x = -1$.

Find the shaded area.



You must use calculus and give the result of any integration needed to solve this problem.

(e) Ben leaves his cup of coffee on the table to cool. The room's temperature remains constant at 18°C.

The rate at which the temperature of the coffee changes at any instant is proportional to the difference between the temperature of the coffee and the room temperature at that instant.

(i) Write the differential equation that expresses this statement.

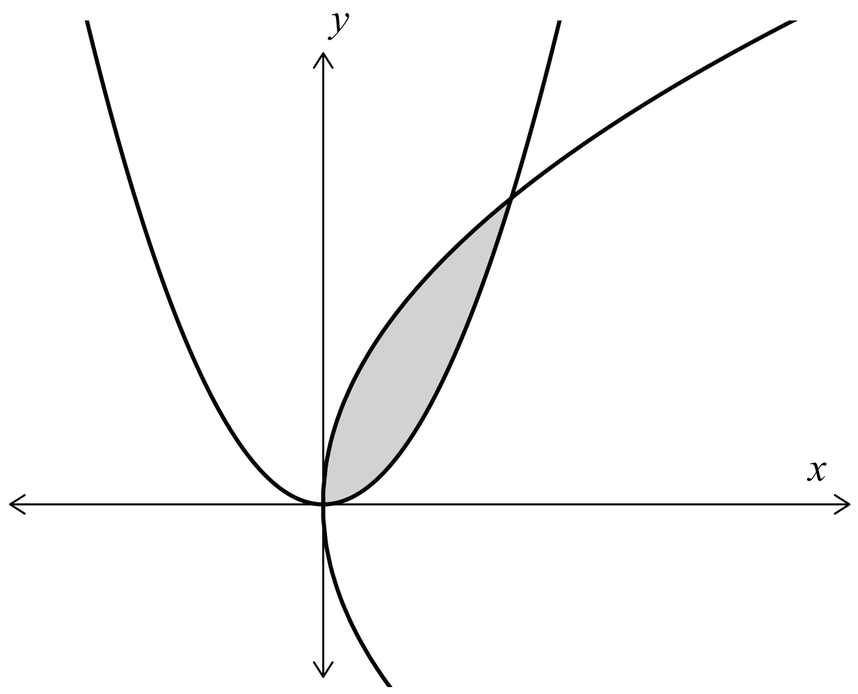
(ii) The temperature of the coffee when it is poured is 65°C.

After 2 minutes, its temperature is 62°C.

Ben can drink the coffee when its temperature is 54°C.

Solve the differential equation in (i) to find how long from when it was poured, that Ben has to wait before he can first drink the coffee.

(e) The diagram below shows the graphs of the curves $y^2 = px$ and $y = px^2$, where $p > 1$.



Show that the area between the two curves is $\frac{1}{3}$.

You must use calculus and give the results of any integration needed to solve this problem.

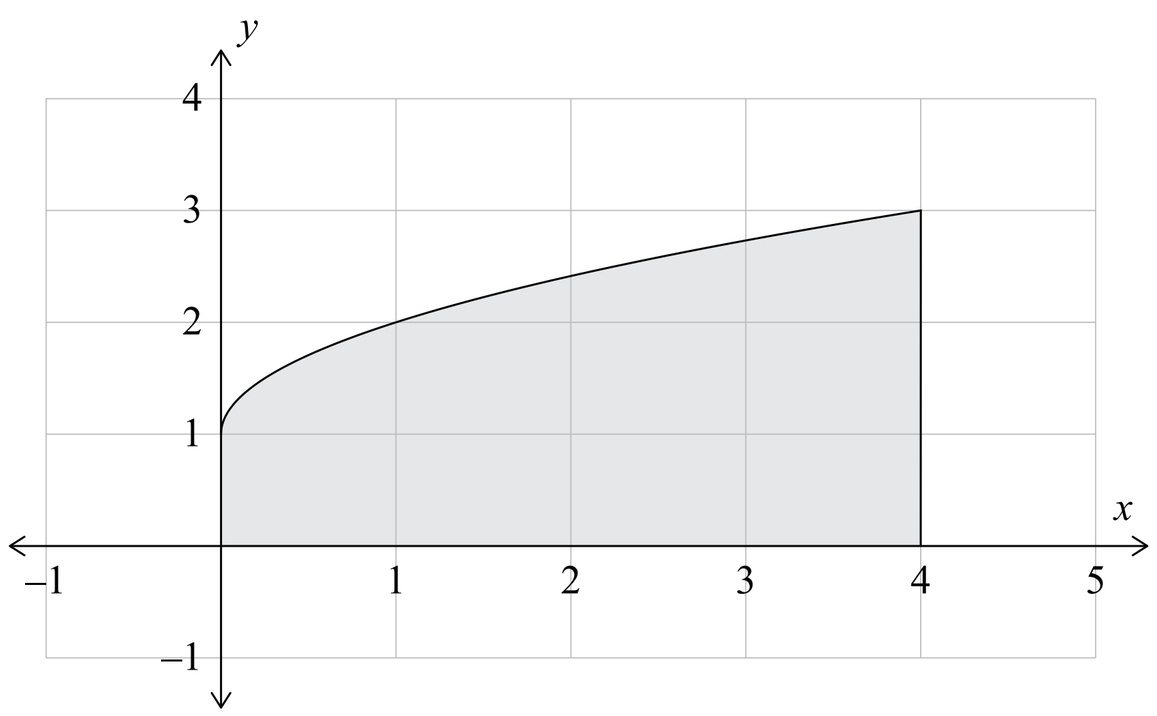
(e) The centre of mass of an object is called the centroid. For a uniformly thin object, the centroid is at $\left(\bar{x}, \bar{y}\right)$ where

$$\bar{x} = \frac{1}{A}\int_a^b x f\left(x\right)\mathrm{d}x \quad \text{and} \quad \bar{y} = \frac{1}{A}\int_a^b \frac{\left[f\left(x\right)\right]^2}{2}\mathrm{d}x$$

A = area of object
a and b are the lower and upper limits of x respectively.

The shape shown shaded in the diagram below is bounded by part of the curve $y = \sqrt{x} + 1$ and the lines $x = 0$, $x = 4$, and $y = 0$.

Find the coordinates $\left(\bar{x}, \bar{y}\right)$ of the centroid of the shape.



You must use calculus and give the results of any integration needed to solve this problem.

(e) If a plant is grown at a constant temperature in a glasshouse, then the rate of growth of the plant depends on the length of the day.

The rate of growth is given by the equation

$$\frac{dh}{dt} = k\left(12 + 3\cos\left(\frac{2\pi t}{365}\right)\right)$$

where t is the time measured from the longest day of the year **in days**

h is the height of the plant, in cm

and k is the growth constant, which is different for each plant.

On the longest day of a particular year, a plant has a height of 84 cm.

75 days later the plant has a height of 91 cm.

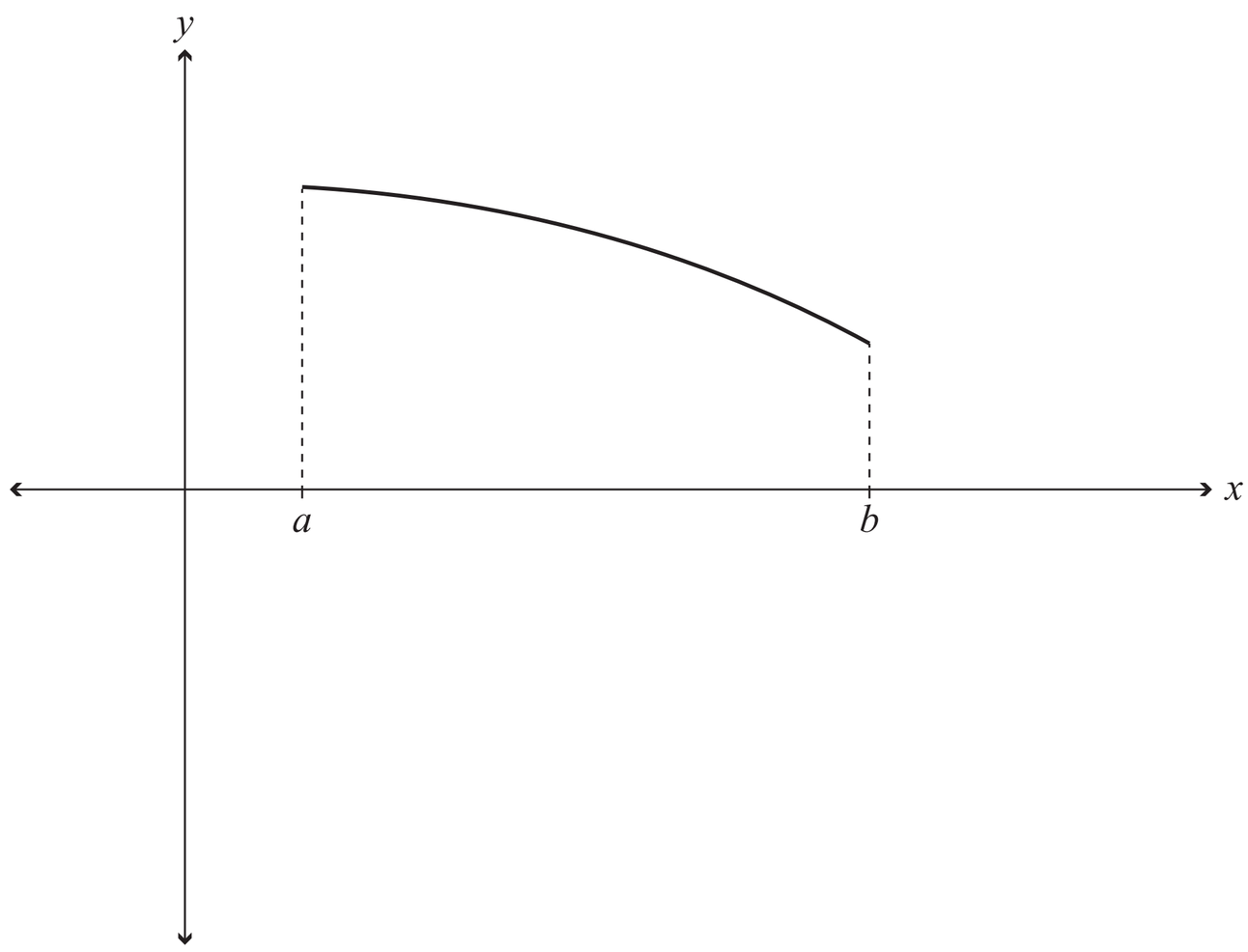
What will the height of the plant be on the longest day of the next year?

Assume the length of a year is 365 days.

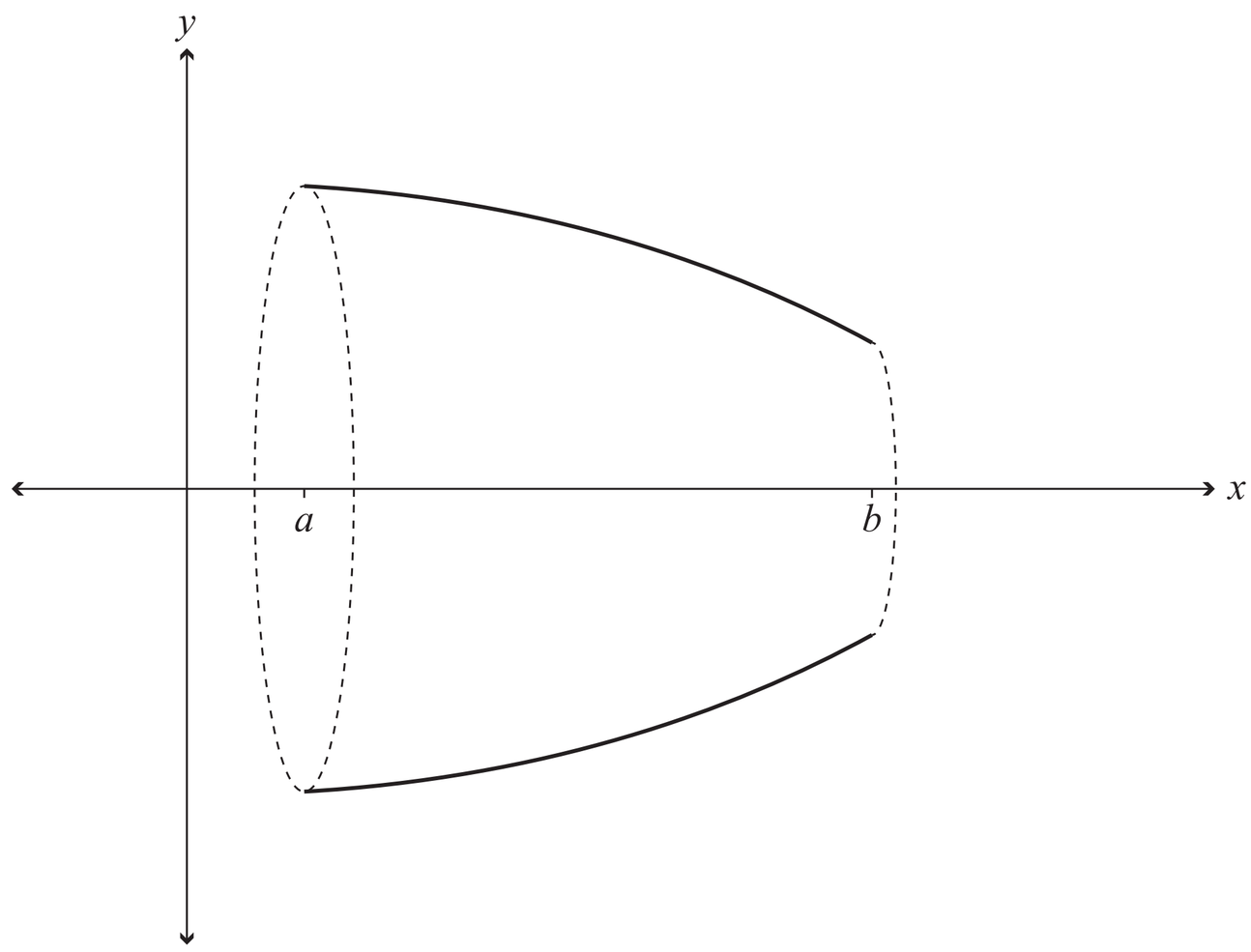
Give the result of any integration needed to solve this problem.

(e)

Consider the curve defined by the function y = $f(x)$, bounded by $x = a$ and $x = b$.



This portion of the curve is rotated around the x-axis, as shown below.

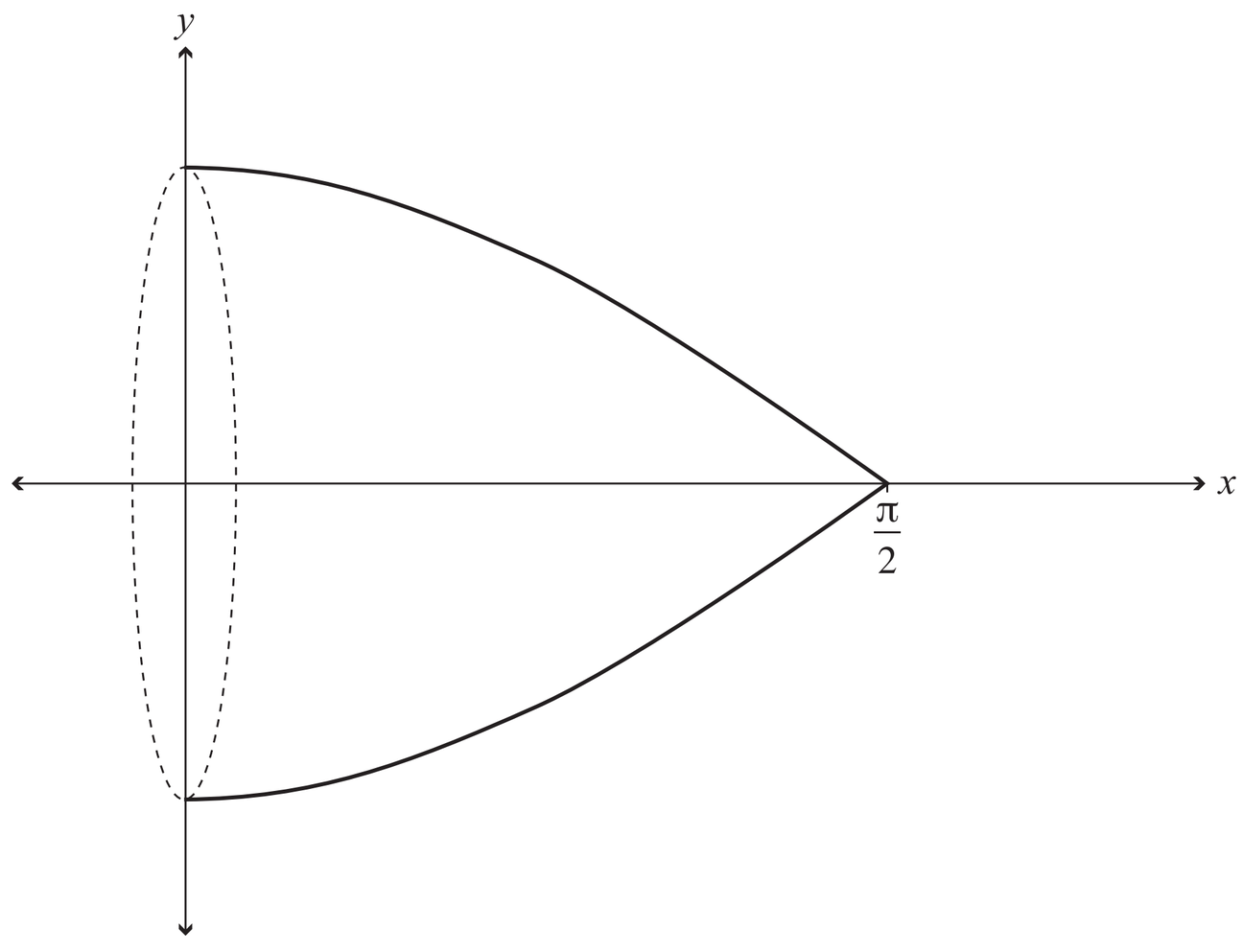


The volume created by this rotation is given by the formula

$$\text{Volume} = \pi \int_a^b \left(f(x) \right)^2 \mathrm{d}x$$

Continued

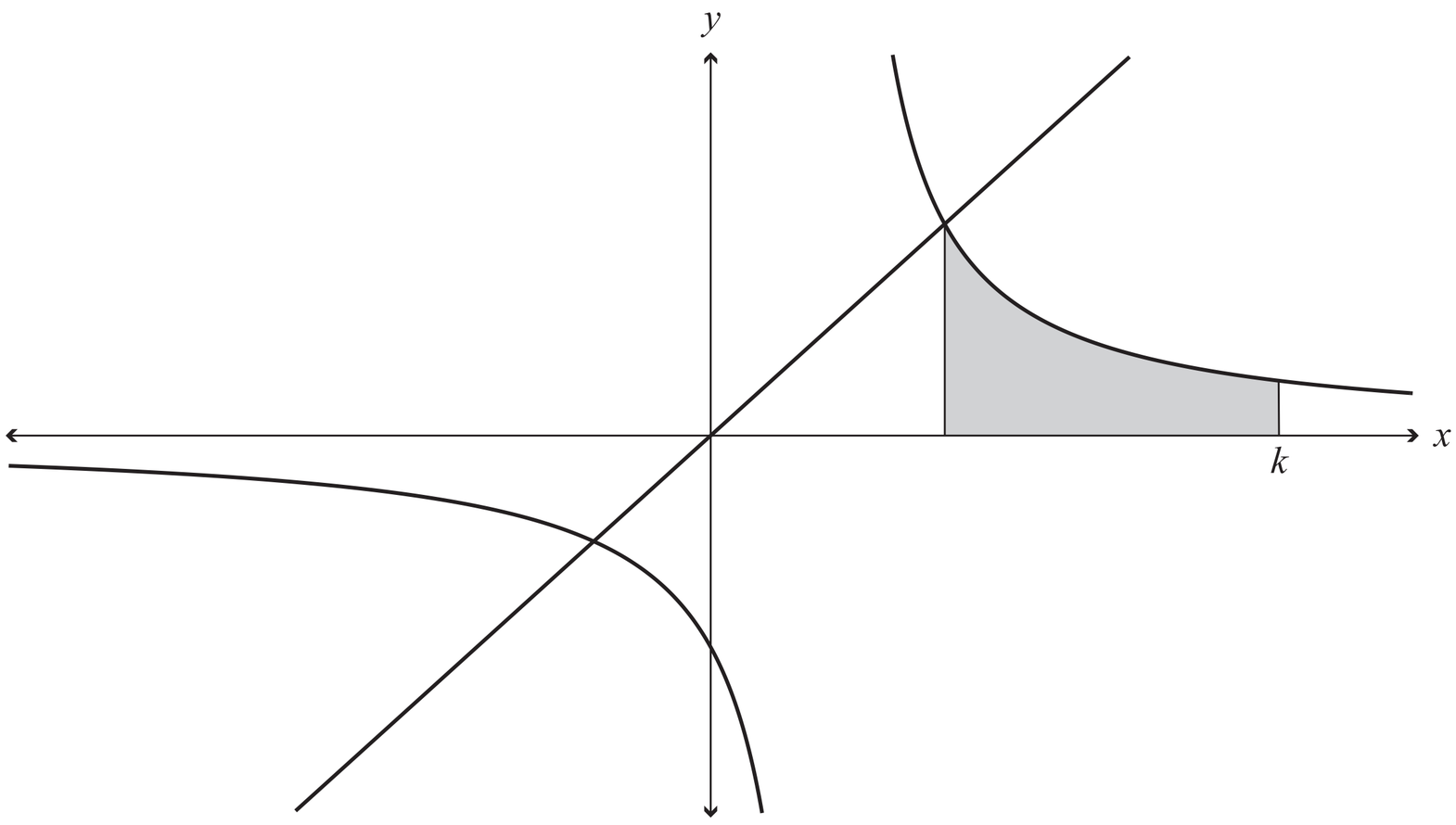
The graph below shows the function $y = \cos x$, between $x = 0$ and $x = \dfrac{\pi}{2}$, rotated around the x-axis.



Find the volume created by this rotation.

You must use calculus and give the results of any integration needed to solve this problem.

(e) The graphs of $y = \dfrac{2}{x-1}$ and $y = x$ are shown on the axes below.



The shaded region has an area of 4 units squared.

Find the value of k.

You must use calculus and give the results of any integration needed to solve this problem.

(e) An object starts from rest.

The object's acceleration is given by the formula $a = B\left(e^{kt}\right)^2$

where a is the acceleration of the object in m s^{-2}

and t is the time, in seconds, from when the object started moving.

Show that the time that it takes the object to reach velocity v_0 is

$$t = \frac{1}{2k}\ln\left(\frac{2v_0k + B}{B}\right)$$

You must use calculus and give the results of any integration needed to solve this problem.

(e) Mr Newton has a container of oil and places it in the garage. Unfortunately, he puts the container on top of a sharp nail and it begins to leak.

The rate of decrease of the volume of oil in the container is given by the differential equation

$$\frac{\mathrm{d}V}{\mathrm{d}t} = -kVt$$

where V is the volume of oil remaining in the container t hours after the container was put in the garage.

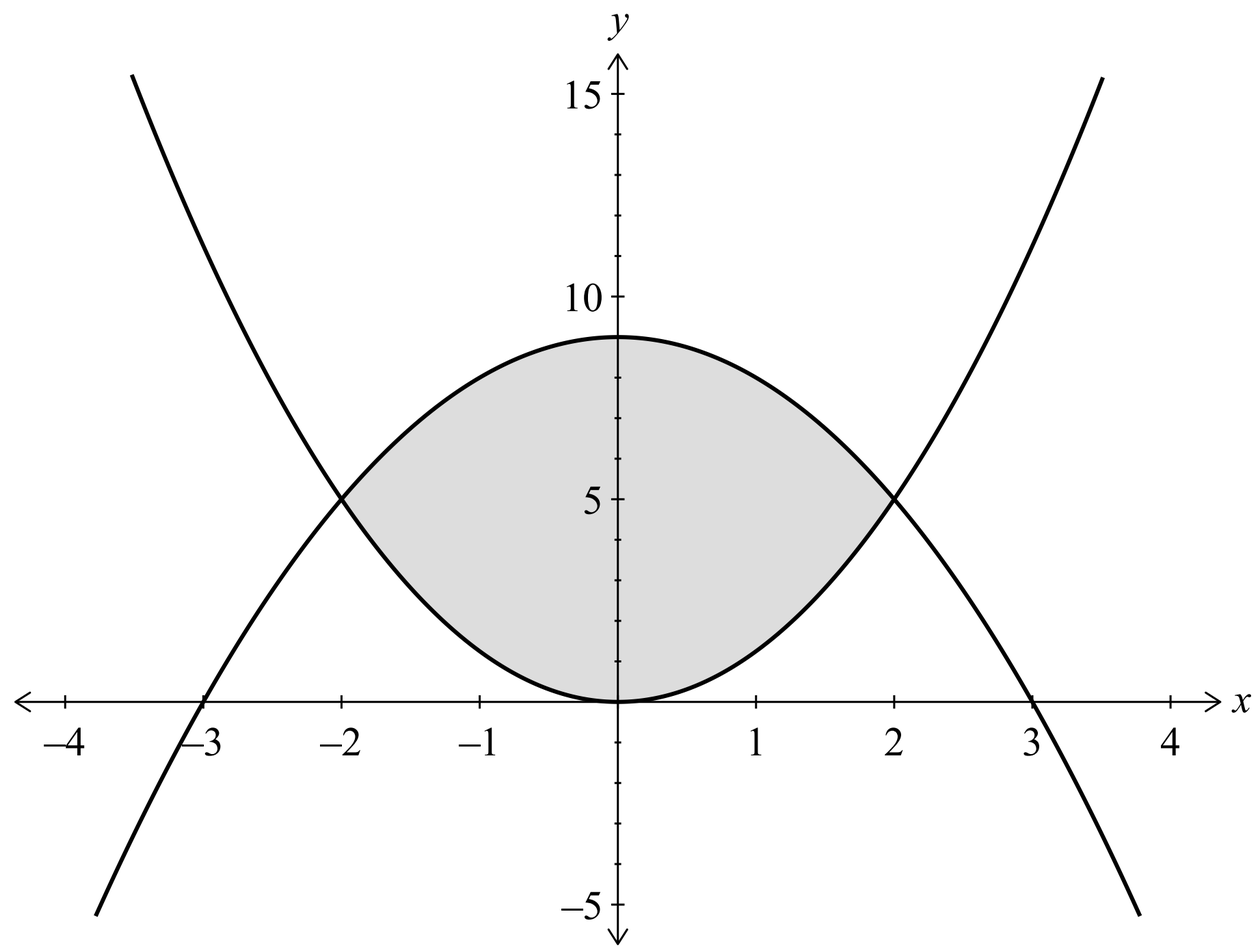
The volume of oil in the container when it was placed in the garage was 3000 mL.

After 20 hours, the volume of oil in the container was 2400 mL.

How much, if any, of the oil will remain in the container 96 hours after it was placed in the garage?

You must use calculus and give the results of any integration needed to solve this problem.

(e) The graphs of $y = (k - 1)x^2$, $k > 1$ and $y = 9 - x^2$ are shown in the diagram below.



The shaded region has an area of 24.

Find the value of k.

You must use calculus and give the results of any integration needed to solve this problem.

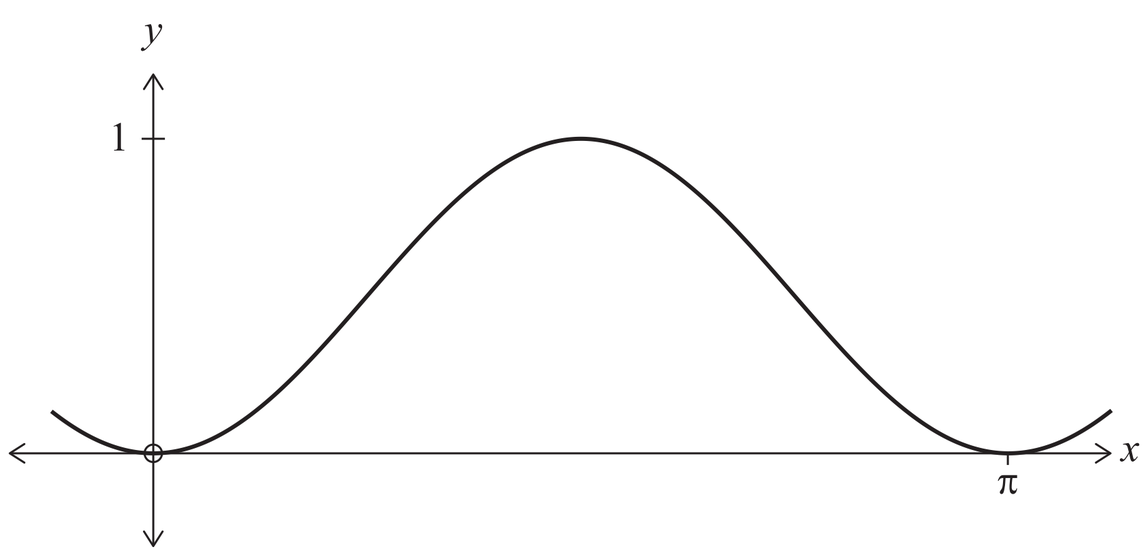
(e) If $\sec x \cdot \dfrac{dy}{dx} = e^{y+\sin x}$ and $y = -1$ when $x = 0$, find the value of y when $x = \dfrac{\pi}{2}$.

(e) The mean value of a function $y = f(x)$ from $x = a$ to $x = b$ is given by

$$\text{Mean value} = \frac{\int_a^b f(x)\,\mathrm{d}x}{b-a}$$

Find the mean value of $y = \sin^2 x$ between $x = 0$ and $x = \pi$.

Part of the graph of $y = \sin^2 x$ is shown below.



You must use calculus and show the results of any integration needed to solve the problem.

(e) The acceleration of an object is modelled by the function $a(t) = \dfrac{20 \ln t}{t}$.

where a is the acceleration of the object in m s^{-2}
and t is the time in seconds since the start of the object's motion.

The object was moving with a velocity of 12 m s^{-1} when $t = 4$.

Find the velocity of the object after 10 seconds.

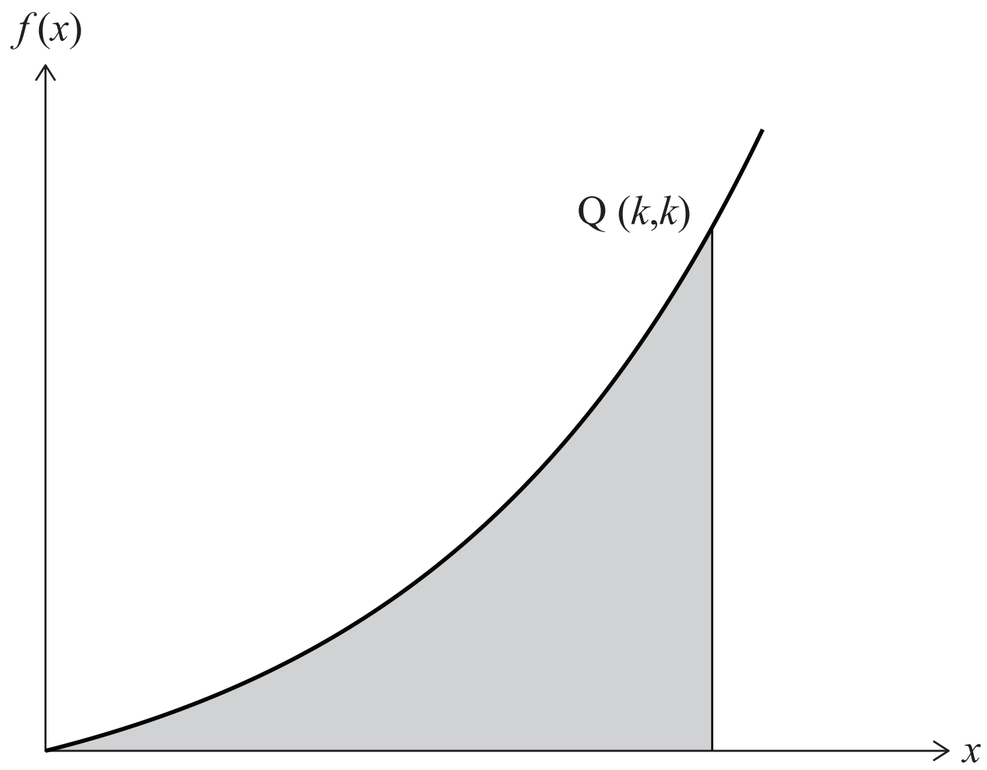
You must use calculus and show the results of any integration needed to solve the problem.

(e) y and t satisfy the differential equation $\frac{dy}{dt} = k\cos 0.5t \times e^{\sin 0.5t}$, $0 \le t \le 5$.

Given that when $t = 0$, $y = 8$, and that when $t = 2$, $y = 12$, find the value of y when $t = 5$.

You must use calculus and show the results of any integration needed to solve the problem.

(e) The diagram below shows the graph of the function $f(x) = \frac{1}{2}\left(e^x - 1\right)$.



The point Q (k,k) lies on the curve.
The shaded region in the diagram is bounded by the curve, the x -axis and the line $x = k$.

Show that the shaded region has an area of $\frac{1}{2}k$.

You must use calculus and show the results of any integration needed to solve the problem.

(e) The mass, m grams, of a burning candle t hours after it was first lit can be modelled by the differential equation

$$\frac{\mathrm{d}m}{\mathrm{d}t} = -k(m-10) \text{ where } k > 0 \text{ and } m \geq 10.$$

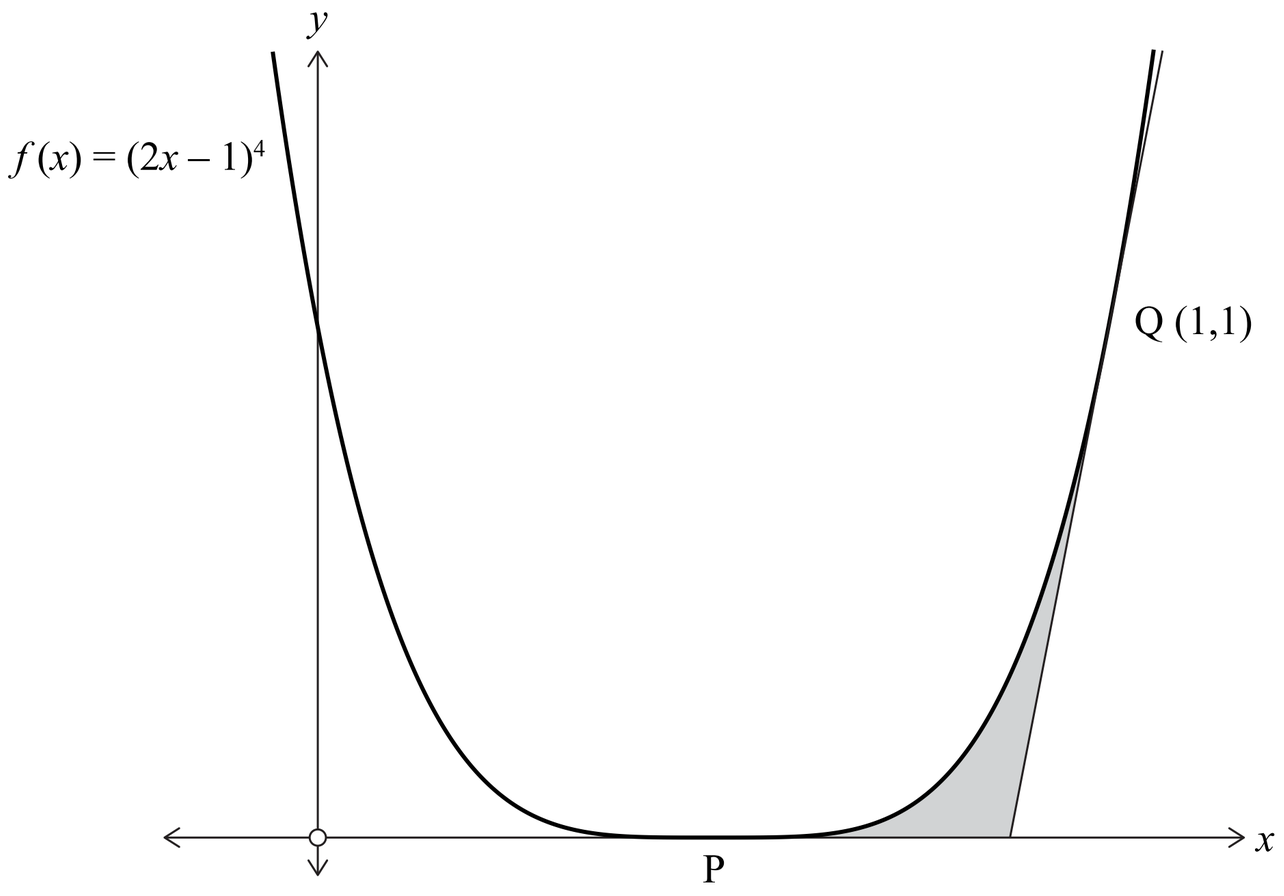
The initial mass of the candle was 140 grams.

3 hours later the mass of the candle had halved.

Find the length of time it will take for the mass of the candle to reduce to 50 grams.

You must use calculus and show the results of any integration needed to solve the problem.

(e) The diagram below shows the graph of the function $f(x) = (2x - 1)^4$.



The curve meets the x-axis at P and the line on the graph is a tangent to the curve at the point Q (1,1).

Find the area of the region bounded by the curve, the x-axis, and the tangent to the curve at Q (shown shaded in the diagram).

You must use calculus and show the results of any integration needed to solve the problem.

(e) The rate of change of quantity N at any instant is given by the differential equation:

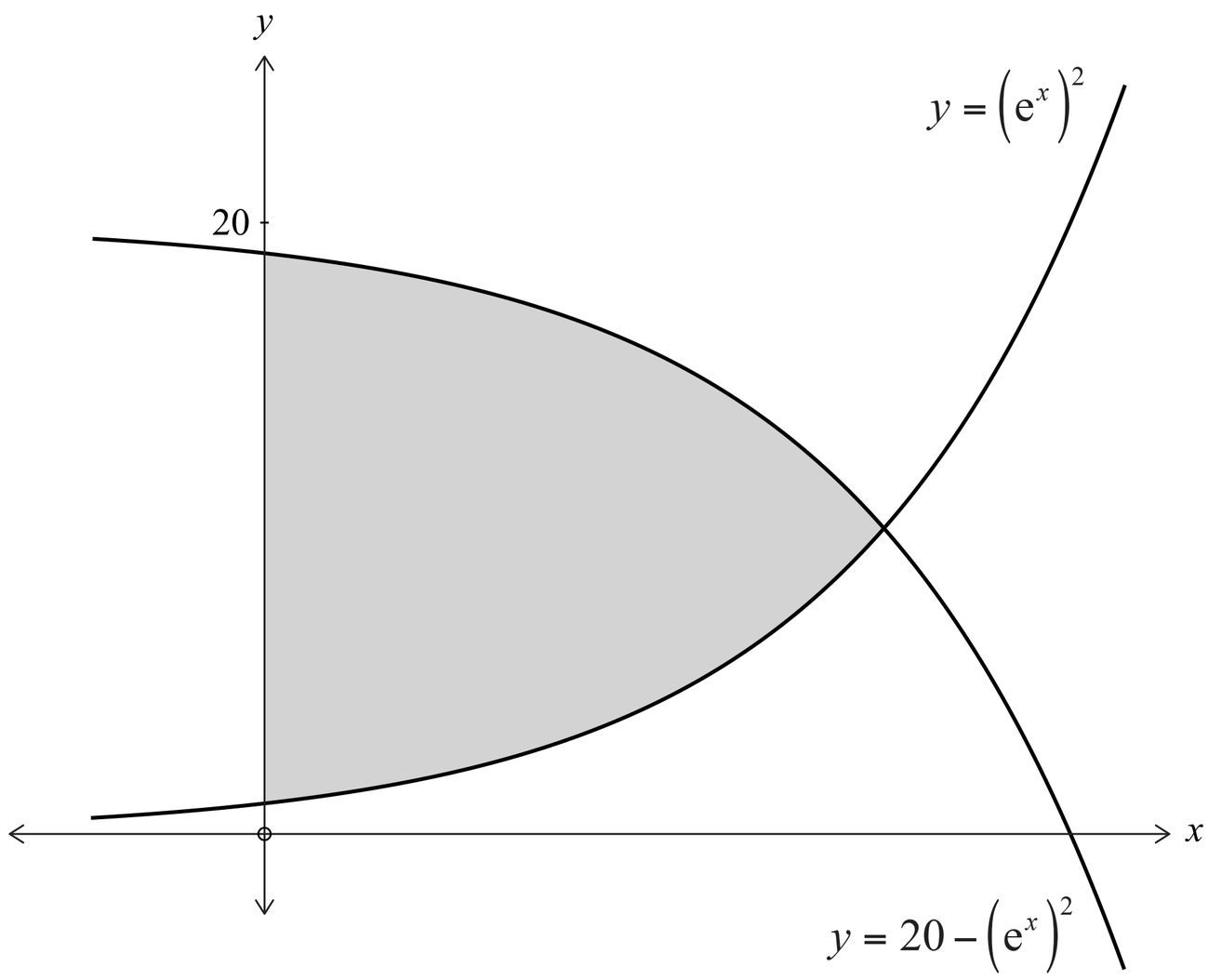
$$\frac{\mathrm{d}N}{\mathrm{d}t} = kN$$

If N has positive values N_1 and N_2 at times t_1 and $2t_1$ respectively, prove that

$$k = \frac{1}{t_1}\ln\left(\frac{N_2}{N_1}\right)$$

You must use calculus and show the results of any integration needed to solve the problem.

(e) The diagram below shows the graphs of the functions $y = \left(e^x\right)^2$ and $y = 20 - \left(e^x\right)^2$.

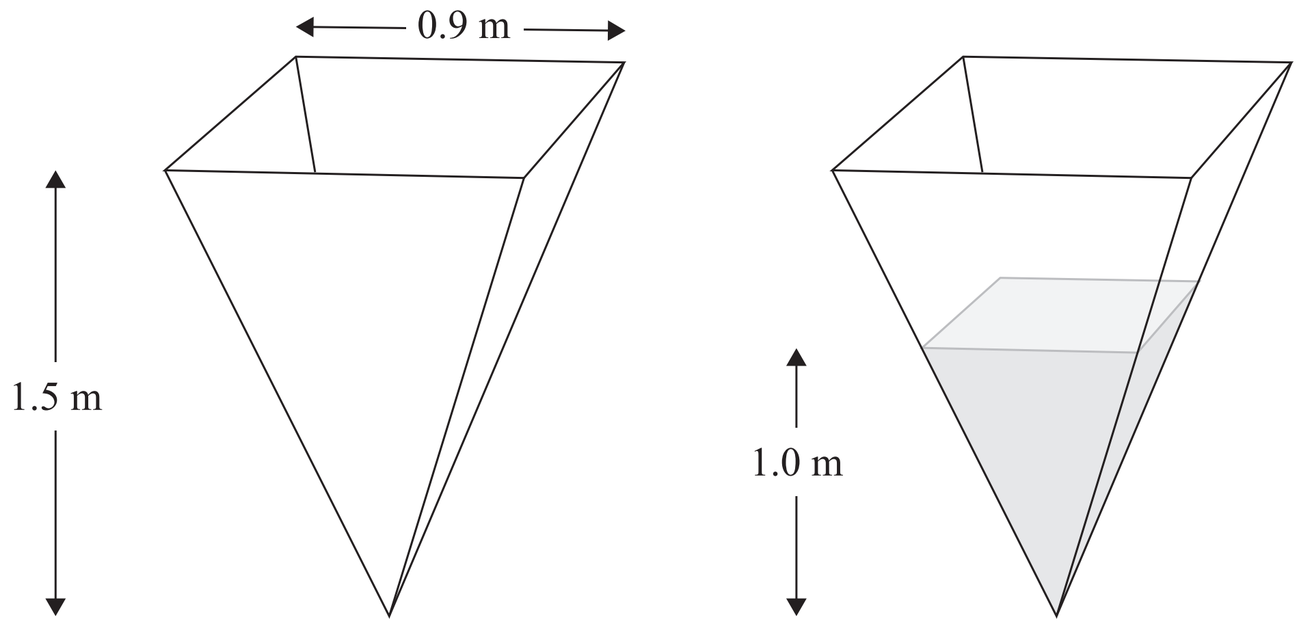


Find the area of the region shaded in the diagram.

You must use calculus and show the results of any integration needed to solve the problem.

(e) An inverted right pyramid has a square base and a height of 1.5 m.

The pyramid is initially filled with water to a depth of 1 m, as shown.



The energy required to pump water out of a tank of height H is given by:

$$E = 9800 \int_{H-d}^{H} (H-h)A(h)\,\mathrm{d}h$$

where E is the energy in joules
d is the initial depth of the water in the tank
h is the depth of the water in the tank at any instant
$A(h)$ is the area of the surface of the water at this instant.

Find the energy required to pump the water out of the tank shown.

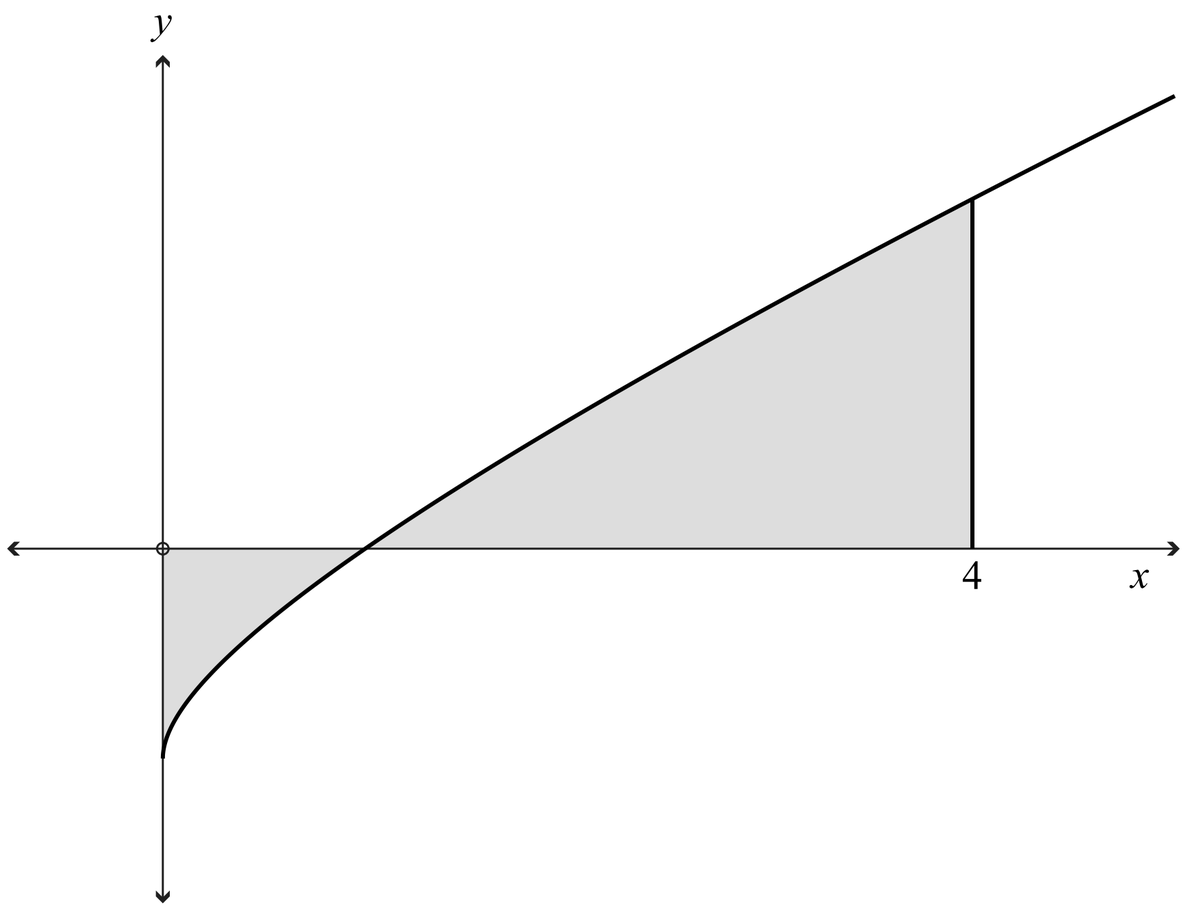
You must use calculus and show the results of any integration needed to solve the problem.

(e) Consider the differential equation $\tan x \dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{\sec^2 x}{y}, 0 < x < \dfrac{\pi}{2}$.

Given that $y = 2$ when $x = \dfrac{\pi}{4}$, find the value(s) of y when $x = \dfrac{\pi}{3}$.

You must use calculus and give the results of any integration needed to solve this problem.

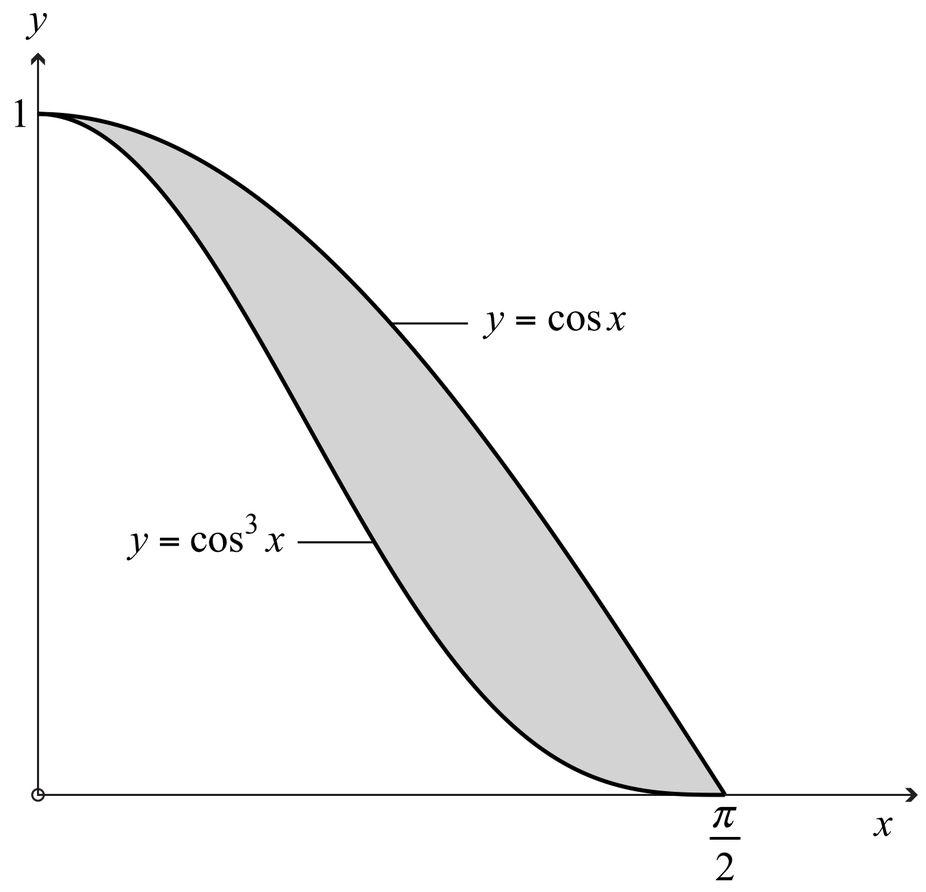
(e) The graph below shows the curve $y = x + 2\sqrt{x} - 3$.



Find the shaded area.

You must use calculus and give the results of any integration needed to solve this problem.

(e) The graph below shows the curves $y = \cos x$ and $y = \cos^3 x$ for $0 \le x \le \frac{\pi}{2}$.



Find the shaded area.

You must use calculus and give the results of any integration needed to solve this problem.

(d) A water tank developed a leak.

6 hours after the tank started to leak, the volume of water in the tank was 400 litres.

10 hours after the tank started to leak, the volume of water in the tank was 256 litres.

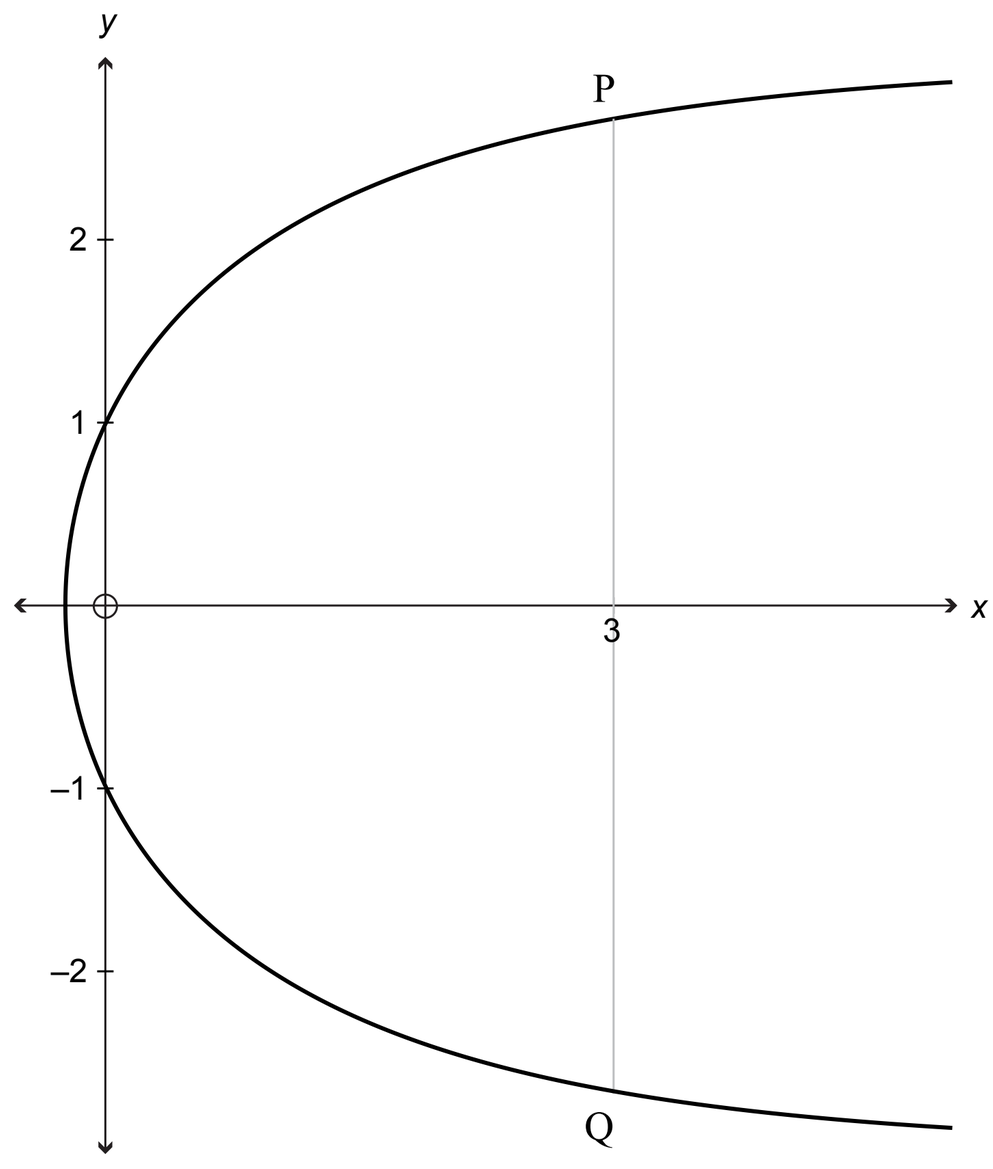
The rate at which the water leaks out of the tank at any instant is proportional to the square root o the volume of the water in the tank at that instant.

How much water was in the tank at the instant it started to leak?

You must use calculus and show the results of any integration needed to solve the problem.

(e) The diagram below shows the graph of a curve $y = f(x)$,

which satisfies the differential equation $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{2}{y\mathrm{e}^{0.5x}}$.

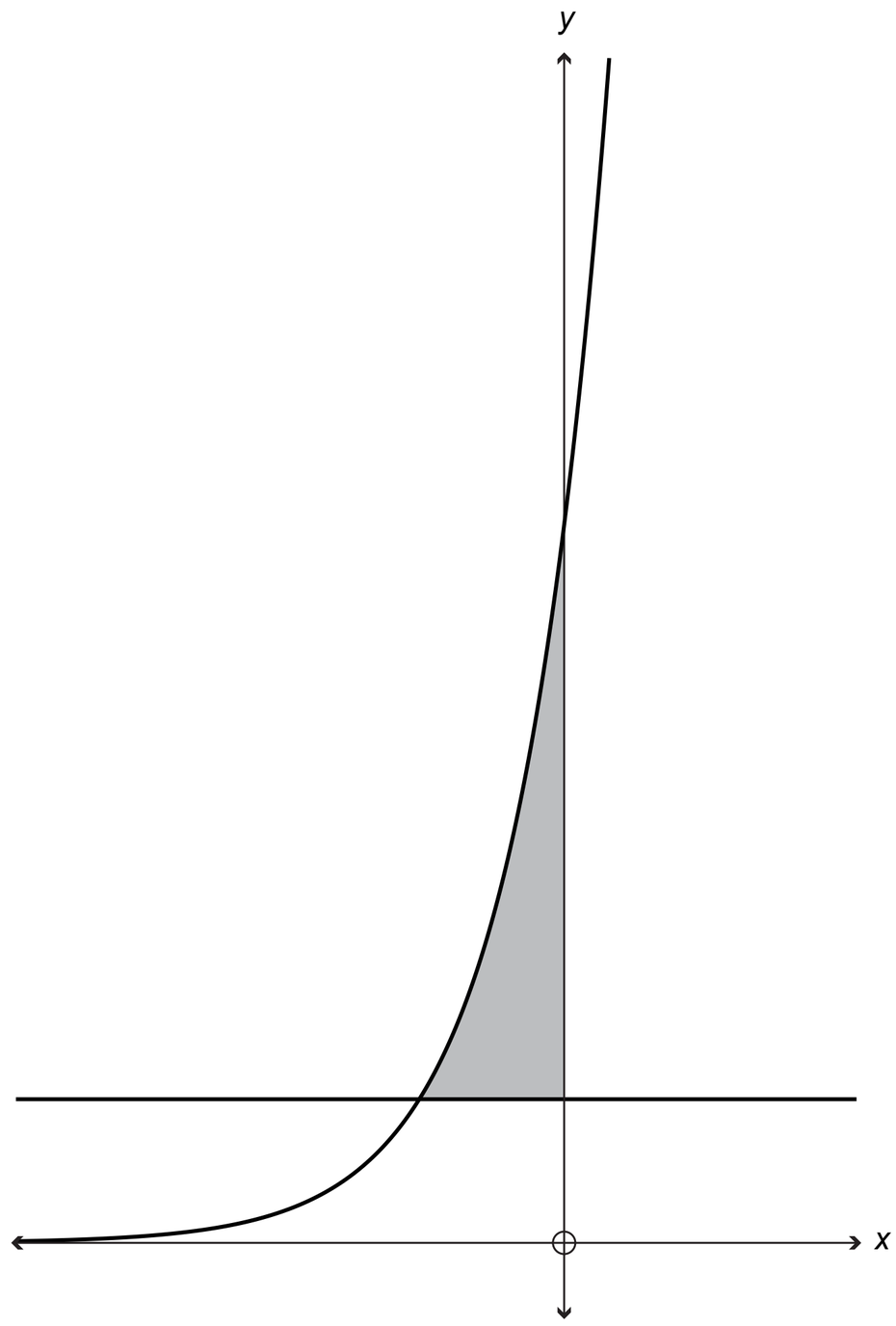


Points P and Q are the points on the graph of the curve that have x-coordinates of 3.

What is the vertical distance between points P and Q?

You must use calculus and show the results of any integration needed to solve the problem.

(e) The graph below shows the functions $y = \left(k\mathrm{e}^{x}\right)^2$ and $y = k$, where k is a constant greater than 1.

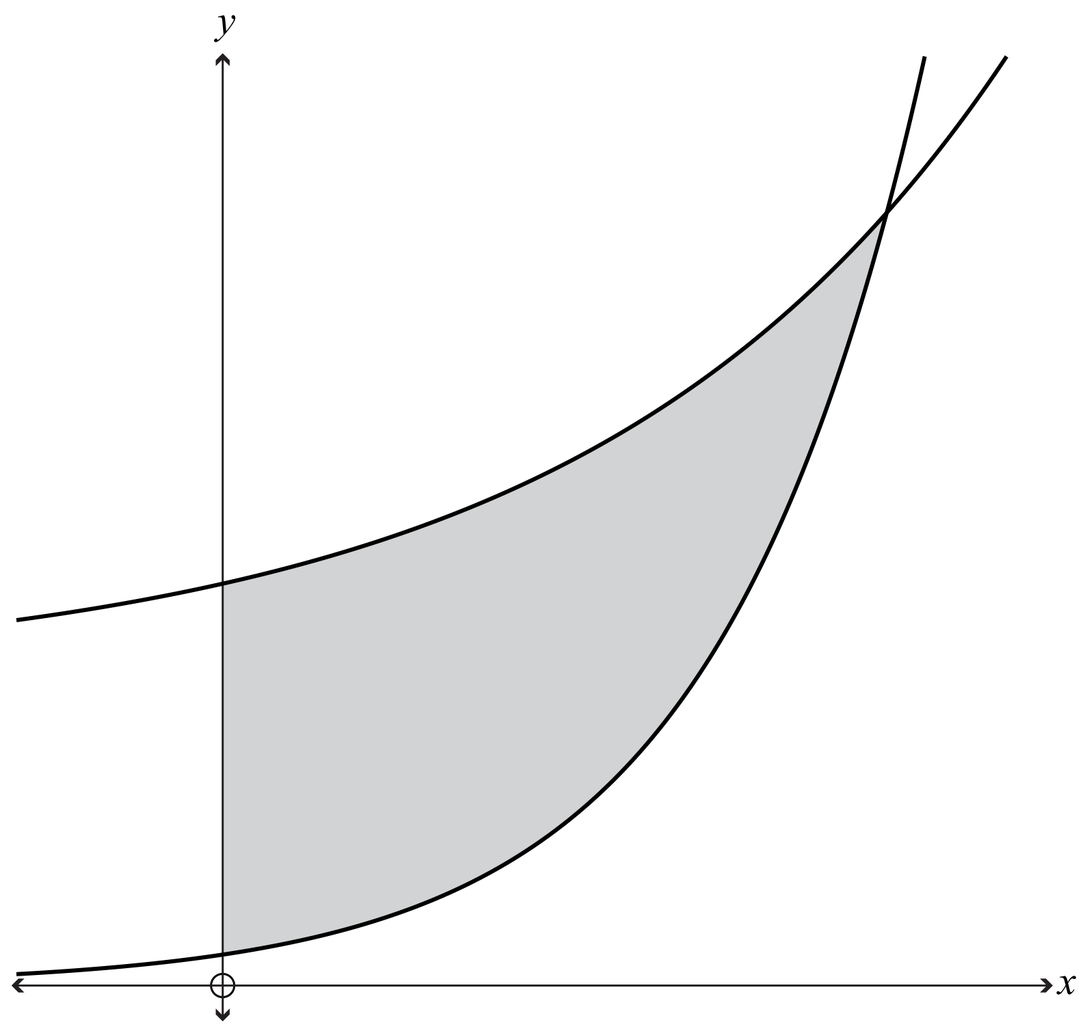


Show that the shaded area is $\frac{k}{2}\left(k - 1 + \ln\frac{1}{k}\right)$.

You must use calculus and show the results of any integration needed to solve the problem.

Clearly show each step of your working.

(e) The graph below shows the functions $y = (e^x)^2$ and $y = 3e^x + 10$.



Find the exact value of the shaded area.

You must use calculus and show the results of any integration needed to solve the problem.

(e) A cylindrical tank of height 150 cm is originally full of oil.

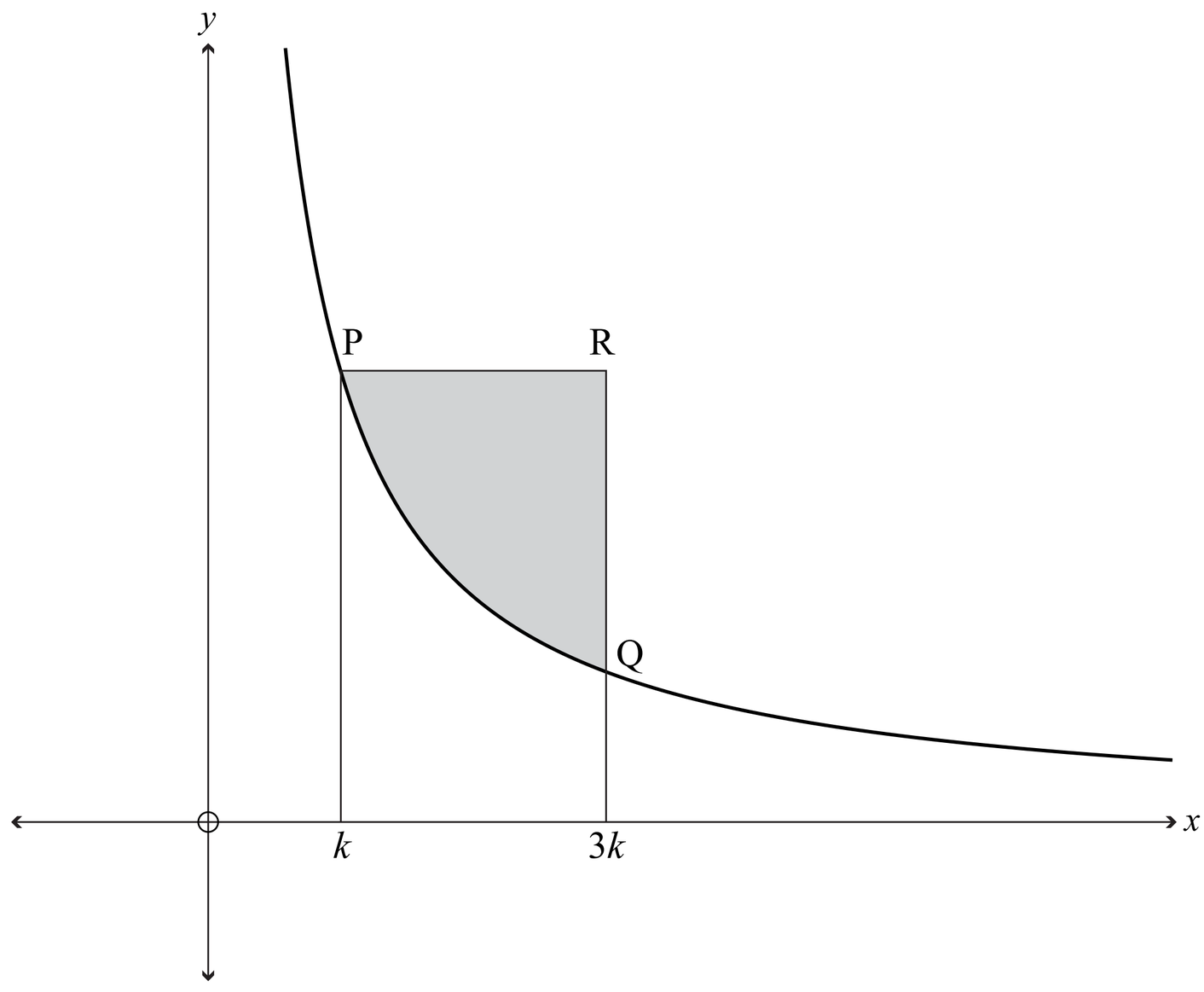
The tank starts to leak out of a hole in its side.

The height h, in cm, of the oil left in the tank after it has been leaking for t minutes can be model by the differential equation $\frac{\mathrm{d}h}{\mathrm{d}t} = \frac{-1}{4}\sqrt{(h-6)^3}$.

Find how long it takes for the oil in the tank to be 15 cm above the bottom of the tank.

You must use calculus and show the results of any integration needed to solve the problem.

(e) The graph below shows part of the curve given by the equation $y = \frac{2}{x}$.



Points P and Q lie on the curve with x-coordinates k and $3k$ respectively, where $k > 0$.

Point R is such that PR is parallel to the x-axis and QR is parallel to the y-axis.

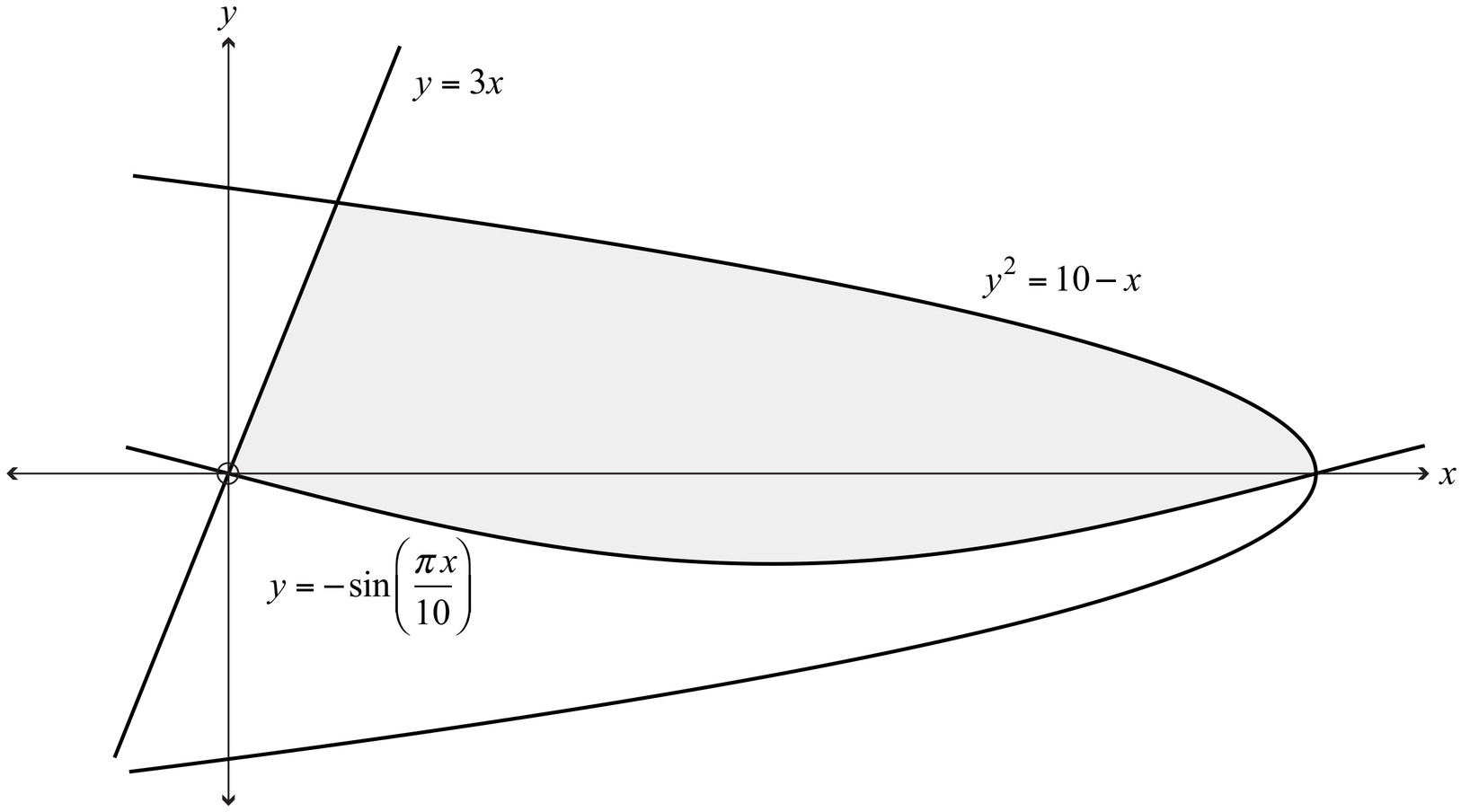
The shaded area can be written in the form $a + b \ln c$, where a, b, and c are integers.

Find the values of a, b, and c.

You must use calculus and show the results of any integration needed to solve the problem.

(e) The shaded region in the diagram below is bounded by the three graphs:

$$y^2 = 10 - x$$
$$y = 3x$$
$$y = -\sin\left(\frac{\pi x}{10}\right)$$



Find the area of the shaded region.

You must use calculus and show the results of any integration needed to solve the problem.

(e) A cake factory has a container of liquid chocolate that is used in the manufacture of chocolate cakes.

The liquid chocolate is pumped out of the container so that the rate of change of the volume of liquid chocolate remaining in the container is proportional to the square of the volume of liquid chocolate remaining.

After one hour of use on a particular day, the volume of chocolate remaining is p litres, where p is a positive constant.

After a further one hour, there are only $\frac{4}{5}p$ litres of chocolate remaining in the container.

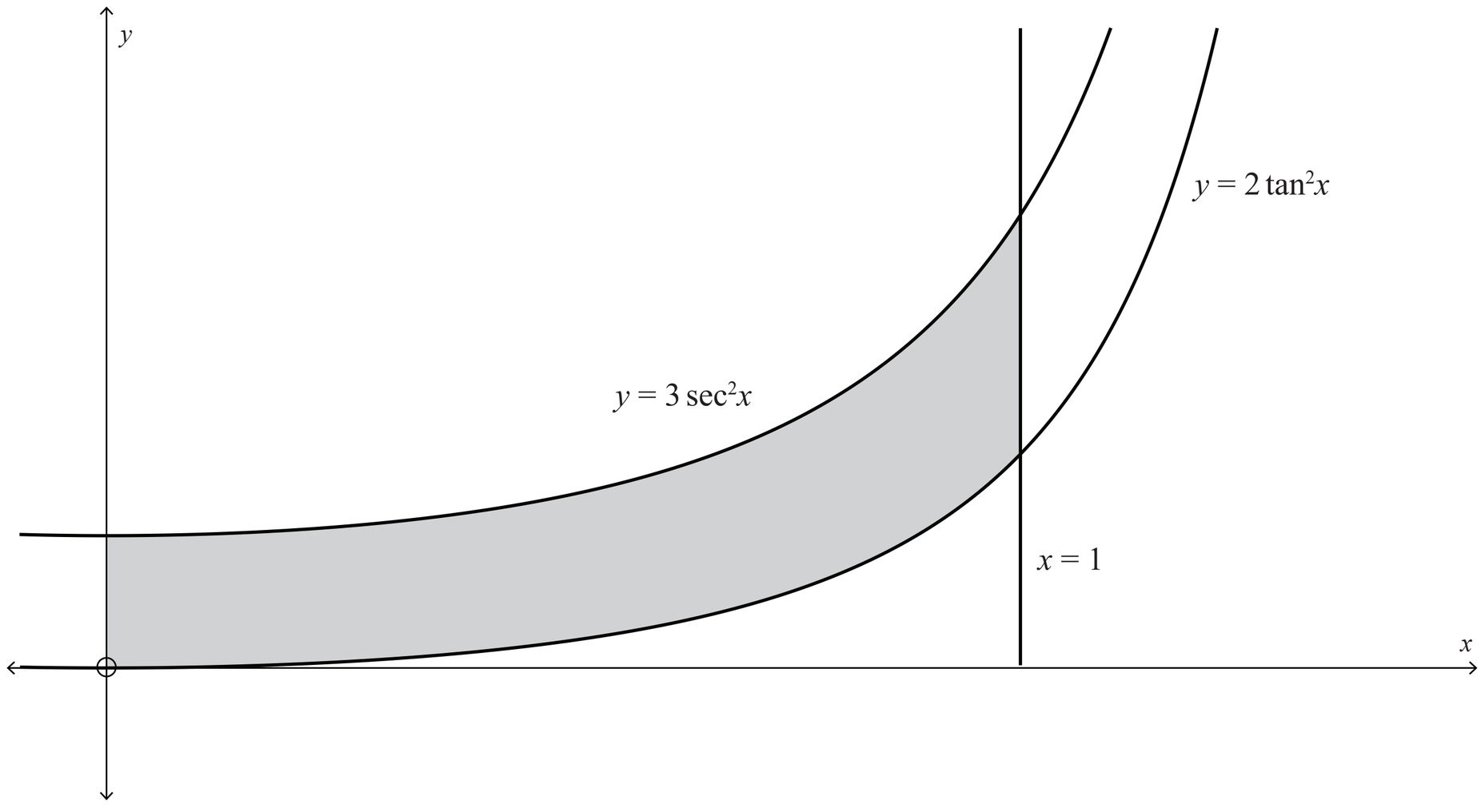
Write a differential equation that models this situation, and solve it to calculate how much liquid chocolate was in the container at the start of the day, giving your answer in terms of p.

(e) Consider the differential equation $\left(1-x^2\right)\left(1+y\right)\dfrac{\mathrm{d}y}{\mathrm{d}x}+\left(1-x\right)\left(1-y^2\right)=0$.

Given that $y = 0$ when $x = 2$, find the value(s) of y when $x = 6$.

You must use calculus and show the results of any integration needed to solve the problem.

(e) The graph below shows the curves $y = 3\sec^2 x$ and $y = 2\tan^2 x$.



Find the area of the shaded region enclosed by the two curves, $x = 1$, and the y-axis.

You must use calculus and show the results of any integration needed to solve the problem.

(e) The mass, M, of a spherical object, with radius p, can be approximated by

$$M = \int_0^p 4\pi r^2 \frac{a}{\left(1 + br^3\right)} \mathrm{d}r$$

where a, b, and p are all positive constants.

Using this formula, find an expression for the mass, M, of a spherical object, giving your answer in terms of a, b, p, and π.

(e) A teacher makes a cup of coffee at the start of interval. The teacher leaves the cup of coffee in the staff room where the temperature is 18 °C.

After 30 minutes, the temperature of the cup of coffee is 50 °C, but the teacher believes that the coffee is still too hot to drink.

When the teacher returns again, after a further one hour, the cup of coffee has cooled down to a temperature of 30 °C.

The rate at which the temperature of the cup of coffee changes at any instant is proportional to the difference between the temperature of the cup of coffee, N, and the temperature of the room.

Write a differential equation that models this situation, and then solve it to calculate the temperature of the cup of coffee when it was made.

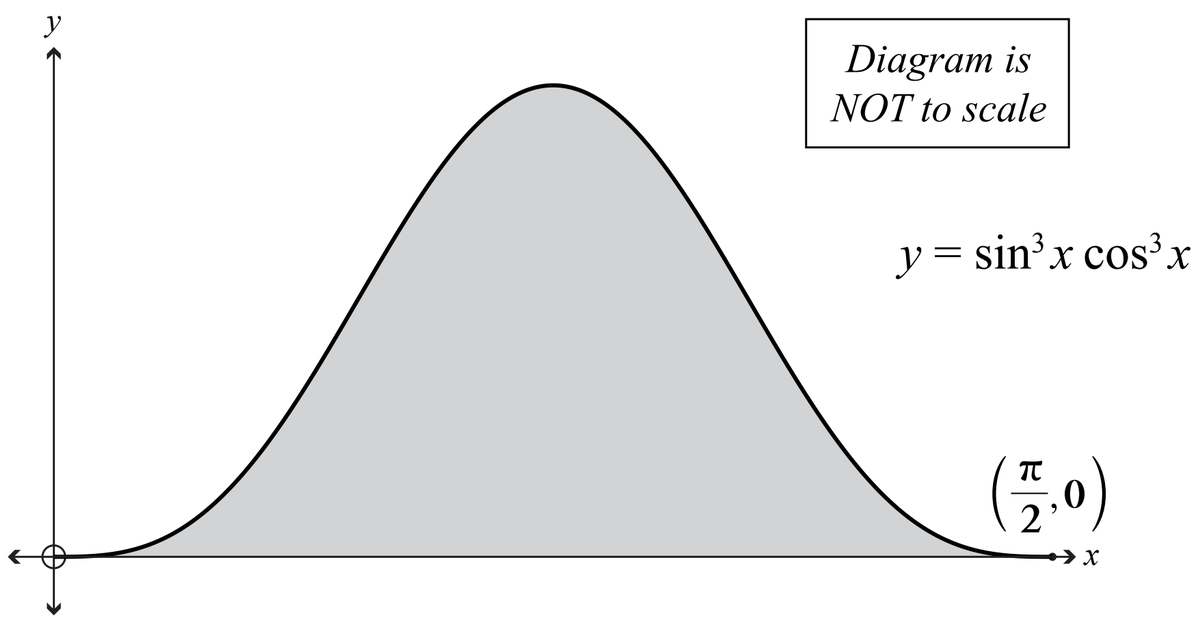
You must use calculus and show the results of any integration needed to solve the problem.

(e) Consider the differential equation $y - xy - (1 + x)\dfrac{dy}{dx} = 0$.

Given that $y = 3$ when $x = 0$, find the value(s) of y when $x = 2$.

You must use calculus and show the results of any integration needed to solve the problem.

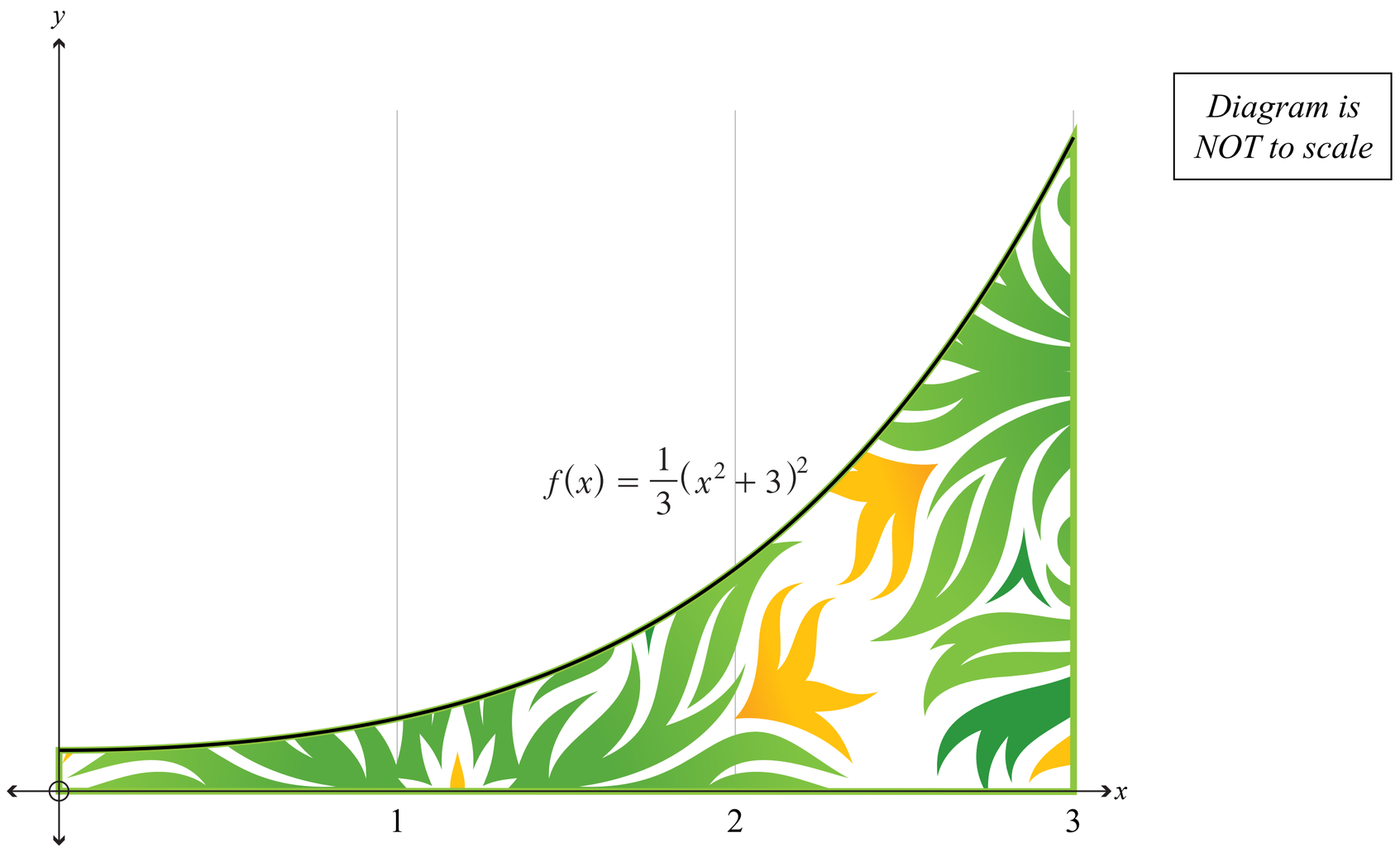
(e) The graph below shows the function $y = \sin^3 x \cos^3 x$.



Find the shaded area under the curve between the values of $x = 0$ and $x = \frac{\pi}{2}$.

You must use calculus and show the results of any integration needed to solve the problem.

(e) Murray is planning to hang a piece of his art on a wall. This is shown in the diagram below.



The equation of the curved edge of the piece of art is $f(x) = \frac{1}{3}(x^2 + 3)^2$.

Murray has researched a way to make the picture balance by using the following formula to find the x-value of the hanging position: $\dfrac{\int_0^3 x\, f(x)\, \mathrm{d}x}{\int_0^3 f(x)\, \mathrm{d}x}$,

where $f(x)$ is the equation of the curved edge of the piece of art.

Use this formula to find the x-value of the hanging point.

You must use calculus and show the results of any integration needed to solve the problem.